



Module Presenter's Manual

for

Java Programming-I

***Effective from: December 2023
Ver. 1.0***

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	December 2023	New	-

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1. Introduction

At the end of this course, students will be able to:

- Develop classes and how declare classes
- Create a Java class
- Declare and initialize variables
- Explore the different data types in Java
- Understand the major operators
- Understand the use of decision making and loop constructs
- Work with Arrays and String classes
- Learn packages and access specifiers
- Use inheritance to declare and define a subclass for a superclass
- Understand nested class
- Describe error handling in a Java program
- Explain the new Date and Time API
- Explain the Functional Programming in Java
- Explain the new Features of Java
- Explain JDK 20 new Features and Functionalities

2. Information on Session Allocation

Module	Online (No. of Hours)	Self-Study (No. of Hours)
Java Programming-I	36	12

Throughout this Presenter's Manual, the module **Java Programming-I** will be referred to as **JAVAP**.

3. Module Deliverables available on OnlineVarsity

To aid the learning process, following are the deliverables:

Student Deliverables:

1. Learner's Guide (eBook)

Resources available on OnlineVarsity for Students:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Glossary - Student can access a list of subject related specialized words with their definitions.
	FAQ - Student can access frequently asked questions and their answers.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

4. Week-wise Session Schedule

- A Session is of 2 hours duration.

➤ Week-Wise Schedule

Week	Day 1	Day 2		Day 3	Day 4	
1	Session 1 JAVAP-TL1	Session 2 JAVAP-TL2		Session 3 JAVAP-TL3	Session 4 JAVAP-TL4	S1
2	Session 5 JAVAP-TL5	Session 6 JAVAP-TL6		Session 7 JAVAP-TL7	Session 8 JAVAP-TL8	S2
3	Session 9 JAVAP-TL9	Session 10 JAVAP-TL10		Session 11 JAVAP-TL11	Session 12 JAVAP-TL12	S3
4	Session 13 JAVAP-TL13	Session 14 JAVAP-TL14		Session 15 JAVAP-TL15	Session 16 JAVAP-TL16	S4
5	Session 17 JAVAP-TL17	Session 18 JAVAP-TL18	S5 & S6			

TL: Online Session

S: Self Study (over and above the module duration)

4. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	JAVAP-TL1	All the topics as listed below from Session 1 of <i>Java Programming - The Complete Guide for Beginners</i> book should be covered in this session. <u>Session 1 – Introduction to Java</u> ➤ Explain structured programming paradigm ➤ Explain object-oriented programming paradigm ➤ Explain features of Java as a OOP language ➤ Describe Java platform and its components ➤ List different editions of Java ➤ Explain evolution of Java Standard Edition (Java SE) ➤ Describe steps for downloading and installing Java Development Kit (JDK)	<u>Java Programming - The Complete Guide for Beginners</u> SG - Session 1 XP - Session 1 TG - Session 1
2	JAVAP-TL2	All the topics as listed below from Session 2 and Session 3 of <i>Java Programming - The Complete Guide for Beginners</i> book should be covered in this session. <u>Session 2 – Variables, Data Types, and Operators</u> ➤ Explain variables and their purpose ➤ Explain the syntax of variable declaration ➤ Explain the rules and conventions for naming variables ➤ Explain data types ➤ Explain primitive and reference data types ➤ Explain escape sequence ➤ Explain format specifiers ➤ Identify and explain different type of operators ➤ Explain the concept of casting ➤ Explain implicit and explicit conversion	<u>Java Programming - The Complete Guide for Beginners</u> SG - Session 2 & 3 XP - Session 2 & 3 TG - Session 2 & 3

Session No.	Session Title	Session Details	Deliverables' Mapping
		<u>Session 3 – Decision-Making Constructs and Loops</u> ➤ List different types of decision-making statements ➤ Explain the if statement and various forms of if statement ➤ Explain switch-case statement ➤ Compare the if-else and switch-case statement ➤ List different types of loops ➤ Explain the while statement and the associated rules ➤ Identify the purpose of the do-while statement ➤ Identify the necessity of the for statement ➤ Describe nested loops ➤ Compare different types of loops ➤ Illustrate the purpose of jump statements ➤ Describe break statement ➤ Describe continue statement	
3	JAVAP-TL3	The Try It Yourself questions of Session 1 to Session 3 of <i>Java Programming - The Complete Guide for Beginners</i> book should be covered in this session.	<u>Java Programming - The Complete Guide for Beginners</u> Session 1-3

Session No.	Session Title	Session Details	Deliverables' Mapping
4	JAVAP-TL4	All the topics as listed below from Session 4 of <i>Java Programming - The Complete Guide for Beginners</i> book should be covered in this session. <u>Session 4 – Classes, Objects, and Methods</u> ➤ Explain process of creation of classes in Java ➤ Explain instantiation of objects in Java ➤ Explain purpose of instance variables and instance methods ➤ Describe constructors and methods ➤ Explain memory management in Java ➤ Explain object initializers ➤ Describe access specifiers and the types of access specifiers ➤ Explain concept of method overloading ➤ Elaborate the use of <i>this</i> keyword	<u>Java Programming - The Complete Guide for Beginners</u> SG - Session 4 XP - Session 4 TG - Session 4
	S1	All the Assignments from Session 1 to Session 4 of <i>Java Programming - The Complete Guide for Beginners</i> course should be covered in this session.	<u>Java Programming - The Complete Guide for Beginners (OnlineVarsity)</u> Session 1-4

Session No.	Session Title	Session Details	Deliverables' Mapping
5	JAVAP-TL5	All the topics as listed below from Session 5 of <i>Java Programming - The Complete Guide for Beginners</i> book should be covered in this session. <u>Session 5 – Arrays and Strings</u> ➤ Describe an array ➤ Explain declaration, initialization, and instantiation of a single-dimensional array ➤ Explain declaration, initialization, and instantiation of a multi-dimensional array ➤ Explain the use of loops to process an array ➤ Describe ArrayList and accessing values from an ArrayList ➤ Describe String and StringBuilder classes ➤ Describe Wrapper classes, autoboxing, and unboxing	<u>Java Programming - The Complete Guide for Beginners</u> SG - Session 5 XP - Session 5 TG - Session 5
6	JAVAP-TL6	The Try It Yourself questions of Session 4 and Session 5 of <i>Java Programming - The Complete Guide for Beginners</i> book should be covered in this session.	<u>Java Programming - The Complete Guide for Beginners</u> Session 4 & 5

Session No.	Session Title	Session Details	Deliverables' Mapping
7	JAVAP-TL7	All the topics as listed below from Session 6 of <i>Java Programming - The Complete Guide for Beginners</i> book should be covered in this session. <u>Session 6 – Modifiers and Packages</u> ➤ Describe field and method modifiers ➤ Explain different types of modifiers ➤ Explain rules and best practices for using field modifiers ➤ Describe class variables ➤ Explain creation of static variables and methods ➤ Describe package and its advantages ➤ Explain creation of user-defined package ➤ Explain creation of .jar files for deployment	<u>Java Programming - The Complete Guide for Beginners</u> SG - Session 6 XP - Session 6 TG - Session 6
8	JAVAP-TL8	The Try It Yourself questions of Session 6 of <i>Java Programming - The Complete Guide for Beginners</i> book should be covered in this session.	<u>Java Programming - The Complete Guide for Beginners</u> Session 6
	S2	All the Assignments from Session 5 and Session 6 of <i>Java Programming - The Complete Guide for Beginners</i> course should be covered in this session.	<u>Java Programming - The Complete Guide for Beginners (OnlineVarsity)</u> Session 5 & 6

Session No.	Session Title	Session Details	Deliverables' Mapping
9	JAVAP-TL9	All the topics as listed below from Session 7 of <i>Java Programming - The Complete Guide for Beginners</i> book should be covered in this session. <u>Session 7 – Inheritance and Polymorphism</u> ➤ Describe inheritance ➤ Explain the types of inheritance ➤ Explain super class and subclass ➤ Explain the use of super keyword ➤ Explain method overriding ➤ Describe Polymorphism ➤ Distinguish type of reference and type of objects ➤ Explain static and dynamic binding ➤ Explain virtual method invocation ➤ Explain the use of abstract keyword	<u>Java Programming - The Complete Guide for Beginners</u> SG - Session 7 XP - Session 7 TG - Session 7
10	JAVAP-TL10	The Try It Yourself questions of Session 7 of <i>Java Programming - The Complete Guide for Beginners</i> book should be covered in this session.	<u>Java Programming - The Complete Guide for Beginners</u> Session 7

Session No.	Session Title	Session Details	Deliverables' Mapping
11	JAVAP-TL11	All the topics as listed below from Session 8 of <i>Java Programming - The Complete Guide for Beginners</i> book should be covered in this session. <u>Session 8 – Interfaces and Nested Classes</u> ➤ Describe interfaces ➤ Illustrate the purpose of interfaces ➤ Explain implementation of multiple interfaces ➤ Describe private methods in interfaces ➤ Define Abstraction ➤ Explain Nested class ➤ Explain Member class ➤ Explain Local class ➤ Elaborate Anonymous class ➤ Outline static nested class	<u>Java Programming - The Complete Guide for Beginners</u> SG - Session 8 XP - Session 8 TG - Session 8
12	JAVAP-TL12	The Try It Yourself questions of Session 8 of <i>Java Programming - The Complete Guide for Beginners</i> book should be covered in this session.	<u>Java Programming - The Complete Guide for Beginners</u> Session 8
	S3	All the Assignments from <i>Session 7</i> and <i>Session 8</i> of <i>Java Programming - The Complete Guide for Beginners</i> course should be covered in this session.	<u>Java Programming - The Complete Guide for Beginners (OnlineVarsity)</u> Session 7 & 8

Session No.	Session Title	Session Details	Deliverables' Mapping
13	JAVAP-TL13	All the topics as listed below from Session 9 of <i>Java Programming - The Complete Guide for Beginners</i> book should be covered in this session. <u>Session 9 – Exceptions</u> ➤ Describe exceptions ➤ Explain types of errors and exceptions ➤ Elaborate the Exception class ➤ Describe exception handling ➤ Explain try-catch block ➤ Explain finally block ➤ Explain execution flow of exceptions ➤ Summarize guidelines for exception handling	<u>Java Programming - The Complete Guide for Beginners</u> SG - Session 9 XP - Session 9 TG - Session 9
14	JAVAP-TL14	All the topics as listed below from Session 10 of <i>Java Programming - The Complete Guide for Beginners</i> book should be covered in this session. <u>Session 10 – Date and Time API</u> ➤ Explain classes of the Date and Time API ➤ Explain Enum and Clock types ➤ Describe the role of time-zones ➤ Explain support for backward compatibility in the new API ➤ Explain about Stream of Dates	<u>Java Programming - The Complete Guide for Beginners</u> SG - Session 10 XP - Session 10 TG - Session 10
15	JAVAP-TL15	The Try It Yourself questions of Session 9 and Session 10 of <i>Java Programming - The Complete Guide for Beginners</i> book should be covered in this session.	<u>Java Programming - The Complete Guide for Beginners</u> Session 9 & 10

Session No.	Session Title	Session Details	Deliverables' Mapping
16	JAVAP-TL16	All the topics as listed below from Session 11 of <i>Java Programming - The Complete Guide for Beginners</i> book should be covered in this session. <u>Session 11 – Additional Features of Java</u> ➤ Explain Lambda Expressions ➤ Explain the use of Generics ➤ Elaborate the use of annotations ➤ Describe streams ➤ Define modules ➤ Explain switch expressions ➤ Describe <i>text</i> blocks ➤ Outline the use of pattern matching for <i>instanceof</i>	<u>Java Programming - The Complete Guide for Beginners</u> SG - Session 11 XP - Session 11 TG - Session 11
	S4	All the Assignments from Session 9 to Session 11 of <i>Java Programming - The Complete Guide for Beginners</i> course should be covered in this session.	<u>Java Programming - The Complete Guide for Beginners (OnlineVarsity)</u> Session 9-11
17	JAVAP-TL17	All the topics as listed below from Session 12 of <i>Java Programming - The Complete Guide for Beginners</i> book should be covered in this session. <u>Session 12 – JDK 20 New Features and Functionalities</u> ➤ Explain Virtual Threads ➤ Explain Vector API ➤ Describe Structured Concurrency ➤ Outline Scoped Values ➤ Describe Foreign Function and Memory API ➤ Summarize Features of Record Patterns ➤ Explain Pattern Matching for switch Statements and Expressions	<u>Java Programming - The Complete Guide for Beginners</u> SG - Session 12 XP - Session 12 TG - Session 12

Session No.	Session Title	Session Details	Deliverables' Mapping
18	JAVAP-TL18	The Try It Yourself questions of Session 11 and Session 12 of <i>Java Programming - The Complete Guide for Beginners</i> book should be covered in this session.	<u>Java Programming - The Complete Guide for Beginners</u> Session 11 & 12
	S5 & S6	All the Assignments from Session 12 of <i>Java Programming - The Complete Guide for Beginners</i> course should be covered in this session.	<u>Java Programming - The Complete Guide for Beginners (OnlineVarsity)</u> Session 12

6. Library References

➤ Head First Java by Kathy Sierra & Bert Bates
➤ Java: A Beginner's Guide by Herbert Schildt
➤ Java for Dummies by Barry A. Burd

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